YANG ZHANG

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PROFILE

I'm currently a Ph.D. student at École Polytechnique Fédérale de Lausanne, under the supervision of Prof. Zsolt Patakfalvi. My interest is in algebraic geometry in positive characteristics.

EDUCATION

Mathematics, Ph.D. [Fall 2025, Now]
École Polytechnique Fédérale de Lausanne

Mathematics, Exchange [Fall 2024, Fall 2025)
École Polytechnique Fédérale de Lausanne

Mathematics, Master of Science [Spring 2023, Fall 2025)
Universität Bonn

Mathematik, Bachelor of Science [Fall 2020, Spring 2023)
Universität Bonn

WRITINGS

A criterion for associated bundles and the weak Beauville–Bogomolov decomposition
arXiv:2608.10219

Automorphisms of odd dimensional $(2,2)$-complete intersections in characteristic 2
arXiv:2601.04109

On split coverings of Calabi-Yau threefolds in positive characteristics
My master thesis. Available on my homepage.

On the bounded negativity conjecture
My bachelor thesis. Available on my homepage.

TEACHING

Tutor of Algebra II - groups Fall 2026
École Polytechnique Fédérale de Lausanne, Lecturer: Prof. Andrei Negut

Tutor of Algebra III - rings and fields Spring 2026
École Polytechnique Fédérale de Lausanne, Lecturer: Dr. Leonid Monin

Tutor of Algebra II - groups Fall 2025
École Polytechnique Fédérale de Lausanne, Lecturer: Prof. Andrei Negut

Tutor of Algebra I (Commutative Algebra) Spring 2024
Universität Bonn, Lecturer: Prof. Dr. Daniel Huybrechts

Tutor of Einführung in die Algebra (Introduction to the Algebra) Fall 2023
Universität Bonn, Lecturer: Prof. Dr. Daniel Huybrechts

Tutor of Algebraic Geometry II Spring 2023
Universität Bonn, Lecturer: Dr. Gebhard Martin

Tutor of Algebraic Geometry I Fall 2022
Universität Bonn, Lecturer: Dr. Gebhard Martin

Tutor of Mathematik für Physiker (Mathematics for phycists) II Spring 2022
Universität Bonn, Lecturer: Dr. Illia Karabash

SEMINAR TALKS

Kunz's Theorem Fall 2024
Technische Universität München
I presented an alternative proof of the Kunz's theorem on the regularity in positive characteristics using perfection. The proof is given by Bhatt and Scholze.

K-Stability Spring 2024
Universität Bonn
My talk involves the construction of the moduli stack of K-semistable Fano varieties by discussing their boundedness and openness.

Compactified Moduli of K3 Spring 2024
Universität Bonn
I gave the first talk which discusses the basic properties of complex K3 surfaces and moduli spaces of polarized K3 of low degrees.

Chow Groups and Motives Fall 2022
Universität Bonn
The content of my talk consists of introducing the intersection product on the chow ring of a variety, proof of the moving lemma and the projection formula, and the compatibility of intersection product and cup product.

Klassiche Algebraische Geometrie(Classical Algebraic Geometry) Fall 2021
Universität Bonn
I gave a half part in each of two talks. The first talk is about defining product of two projective spaces using the Segre map. The second talk presents the classification of cubic curves on the projective plane.

DISTINCTIONS

IMC Maths 2023, Bronze medal.
IMC Maths 2022, Silver medal.
NOIP China 2018, first award.

EXTRAS

Language skills
Chinese [Standard Mandarin, Zhongyuan Mandarin, Shanghainese] (native), German (C1+), English (C1+), French (A2), Japanese (A2).

Accountant of student company AMBIX [Fall 2017, Fall 2018)
I was active in AMBIX, a student party focusing on organizing business imitation contest for middle school students in grade 9 to 12.